

Introduction to Biophysics II: Quantum physics & biology

Lecture 08: Energy transfer in photosynthesis - III

Förster Resonance Energy Transfer:

- Förster Radius R_0
- Energy transfer efficiency
- Förster Radius from absorption and emission spectra
- Modeling FRET kinetics:
 - solving rate equation
 - Monte Carlo simulation

Reading:

Modern Optical Spectroscopy, chapters 4,5, 7

Additional:

Quantum Effects in Biology, chapter 3

1

Förster Resonance Energy Transfer

Last time...

Hamiltonian of a weakly interacting dimer:

$$\tilde{H} = \tilde{H}_1 + \tilde{H}_2 + \tilde{H}_{21}$$

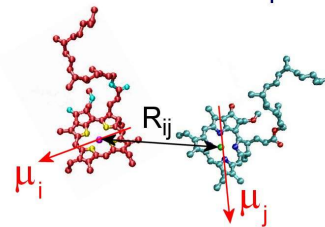
$\uparrow$ $\uparrow$ $\uparrow$
 Hamiltonian of molecule 1 Hamiltonian of molecule 2 Interaction

For constant H_{21}

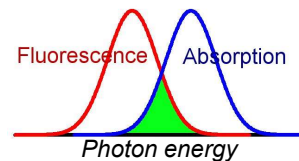
$$k_{rt} = \frac{2\pi}{\hbar} |H_{21}|^2 \int_{-\infty}^{\infty} \rho_{s2}(E_1) \rho_{s1}(E_1) dE_1$$

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(e)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \times \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$

Donor $\rightarrow$ Acceptor



Spectral overlap integral



2

Dipole-dipole interaction

Last time...

$$H_{21} \equiv V_{12} = \left(\frac{f^2}{n^2} \right) \frac{\vec{\mu}_2 \cdot \vec{\mu}_1 - 3(\vec{\mu}_1 \cdot \hat{R}_{21})(\vec{\mu}_2 \cdot \hat{R}_{21})}{R_{21}^3}$$

Donor $\rightarrow$ Acceptor

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_2 \mu_1 \kappa}{R_{21}^3}$$

$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta \leftarrow \text{"orientation factor"}$

$\alpha = 0^\circ$ $\beta = 0^\circ$ $\theta = 0^\circ$ $\kappa = -2$	$\alpha = 0^\circ$ $\beta = -90^\circ$ $\theta = -90^\circ$ $\kappa = 0$	$\alpha = 0^\circ$ $\beta = 90^\circ$ $\theta = 0^\circ$ $\kappa = +1$	$\alpha = 0^\circ$ $\beta = 180^\circ$ $\theta = 180^\circ$ $\kappa = +2$
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3

Point-dipole: energy transfer rate

Last time...

Point-dipole energy transfer rate:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m,u,w} \frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times$$

$$\times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \delta(\Delta E_{1nm} - \Delta E_{2uw})$$

The energy transfer rate $\sim V_{12}^2$.

If a molecule tumbles randomly need to average over all possible κ^2 : $\langle \kappa^2 \rangle = 2/3$

Energy transfer rate is $\sim 1/R^6$ (in point-dipole approximation)

4

Point-dipole: energy transfer rate

Last time...

Point-dipole energy transfer rate:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

Transition dipole strength is often used instead of transition dipole:

$$D \equiv |\vec{\mu}|^2$$

Absorption is proportional to D ; D is usually expressed in Debye² units.

For two identical molecules $\mu_1 = \mu_2 = \mu$:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu^2 \kappa}{R_{21}^3} = \frac{f^2}{n^2} \frac{D \kappa}{R_{21}^3}$$

For D in Debye² and R_{12} in Å:

$$V_{12} = 5.03 \cdot 10^3 \frac{f^2}{n^2} \frac{D \kappa}{R_{21}^3} \quad (V_{12} \text{ in cm}^{-1})$$

Notes:

Typical transition dipole moments vary between 0 and 10 D. (Chl a: ~5-6 D)

1D is the dipole moment of a proton and electron separated by ~0.2 Å

(defined as 1D = 10⁻¹⁸ statcoulomb-centimeters, cgs-esu units)

5

Förster radius

Point-dipole energy transfer rate:

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(ef)}|^2 \sum_{n,m,u,w} \frac{\exp(-E_{n(1)}/kT)}{Z_1} |\langle \chi_{1m} | \chi_{1n} \rangle|^2 \times$$

$$\times \frac{\exp(-E_{u(2)}/kT)}{Z_2} |\langle \chi_{2w} | \chi_{2u} \rangle|^2 \delta(\Delta E_{1nm} - \Delta E_{2uw})$$

It is useful to rewrite as:

$$k_{rt} = \frac{\kappa^2}{\tau} \left(\frac{|R_{21}|}{R_0} \right)^{-6}$$

For tumbling molecules average $\kappa^2 \sim 1$

↑ "Förster radius"
Fluorescence lifetime of the donor
in the absence of the acceptor

Q: What is the rate of energy transfer when $R_{21} = R_0$?

Q: What is the energy transfer quantum efficiency at $R_{21} = R_0$?

6

Energy transfer efficiency

Efficiency of electron transfer:

The fraction of the excitations that transfer from donor to acceptor:

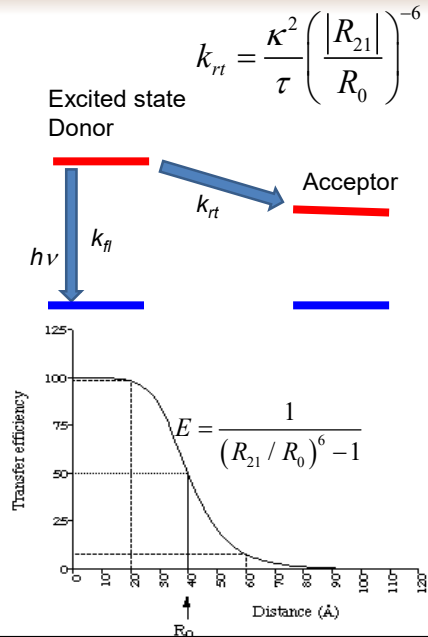
$$E = \frac{k_{rt}}{k_{fl} + k_{rt}} = \frac{1}{k_{fl}/k_{rt} + 1}$$

Efficiency is close to 1 when the energy transfer rate is significantly faster than the decay of the donor's excited state k_{fl} in the absence of energy transfer, which is equal to the fluorescence lifetime

Approximating average $\kappa^2 \sim 1$:

$$E = \frac{1}{\tau^{-1} / \left(\tau^{-1} \{R_{21} / R_0\}^{-6} \right) + 1}$$

$$E = \frac{1}{(R_{21} / R_0)^6 + 1}$$



7

FRET parameters from experiment

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \int_{-\infty}^{+\infty} \rho_{s2}(E) \rho_{s1}(E) dE$$

$$k_{rt} = \frac{2\pi}{\hbar} \frac{f^4}{n^4} \frac{\mu_1^2 \mu_2^2 \kappa}{R_{21}^6} \int_{-\infty}^{+\infty} \rho_{s2}(E) \rho_{s1}(E) dE$$

$$k_{rt} = \tau^{-1} (|R_{21}| / R_0)^{-6}$$

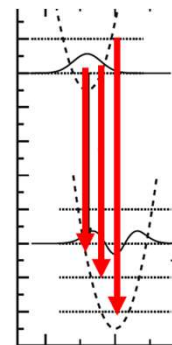
$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

Finding R_0 : from experimental data

For given R_{21} and orientation factor κ , R_0 will depend on the transition dipole moments of donor and acceptor plus nuclear term (in point-dipole approximation)

Can use experimentally measured fluorescence spectrum of a donor and absorption spectrum of acceptor to find R_0 , since these spectra also depend on transition dipole moments.



8

FRET parameters from experiment

Finding R_0 : from experimental data

1. **Donor**: transition dipole moment is related to spontaneous emission rate, Einstein coefficient:

$$1/\tau_{\text{radiative}} = A = (32\pi^3 n f^2 / 3\hbar\lambda^3) D_{ba}$$

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

$$D_{ba} = |\mu_1|^2$$

The faster the rate, the larger the transition dipole strength, $D_{ba} \sim 1/\tau_{\text{radiative}}$. In the presence of competing decay channels, the measured lifetime τ will be shorter:

$$\Phi = \tau/\tau_{\text{radiative}} = \text{fluorescence (quantum) yield}$$

Expect $D_{ba} \sim \Phi/\tau$

Starting from Einstein's coefficient A, can show:

$$D_{ba(1)} X(E) \rho_1(E) = \left(\frac{3\hbar c^3 \Phi}{32\pi^3 n f^2 \tau} \right) \frac{F_1(\nu) \nu^{-3}}{\int F_1(\nu) d\nu}$$

Thermally weighted Franck-Condon factor for emission at energy $E = \hbar\nu$
Density of excited donor states at energy E
F(ν)d ν = Fluorescence probability in interval $\nu \dots \nu + d\nu$
Total fluorescence energy in interval $\nu \dots \nu + d\nu$

...next slide discusses $\tau_r = \tau/\Phi$...

9

Fluorescence yield Φ

An excited state can decay due to different processes:

Radiative decay – with the emission of a photon

Nonradiative decay – everything else, like intersystem crossing into the triplet state, vibrational relaxation energy transfer, etc.

Fluorescence quantum yield = fraction of excitations that result in the emission of light:

$$\Phi = \frac{k_r}{k_{\text{total}}} = \frac{k_r}{k_r + k_n}$$

The lifetime of excited state $\tau = 1/k_{\text{total}}$ is thus shorter than expected for just radiative lifetime $\tau_r = 1/k_r = 1/A$:

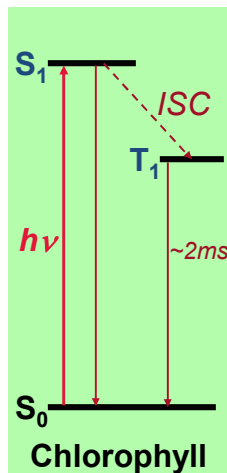
$$k_r = k_{\text{total}} \Phi$$

$$\tau_r = \frac{\tau}{\Phi}$$

Example: Chl a:

$\tau \sim 5$ ns

$\Phi \sim 0.5$



10

FRET parameters from experiment

Finding R_0 : from experimental data

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

2. **Acceptor**: transition dipole moment is related to absorption extinction coefficient and Einstein coefficient B:

$$B_{ba} = \frac{\pi}{3\epsilon_0 \hbar^2} |\mu|^2, \quad R_{b \rightarrow a} = \frac{\pi}{\epsilon_0 \hbar^2} |\mu|^2 \rho(\omega_0)$$

Can start from the transition rate, use the definition of extinction coefficient, and derive:

$$D_{ba(2)} X_2(E) \rho_2(E) = \frac{1}{h} \left(\frac{3000 \ln 10 n h c}{8 \pi^3 f^2 N_A} \right) \frac{\epsilon_2(v)}{v}$$

Thermally weighted
Franck-Condon factor for
acceptor at energy $E=h\nu$
Density of
acceptor states
Avogadro number
Extinction coefficient

...More detailed derivation in textbook...

11

FRET parameters from experiment

Finding R_0 : from experimental data

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_1 \mu_2 \kappa}{R_{21}^3}$$

1. Donor: $D_{ba(1)} X(E) \rho_1(E) = \left(\frac{3 \hbar c^3 \Phi}{32 \pi^3 n f^2 \tau} \right) \frac{F_1(v) v^{-3}}{\int F_1(v) dv}$

2. Acceptor: $D_{ba(2)} X_2(E) \rho_2(E) = \frac{1}{h} \left(\frac{3000 \ln 10 n h c}{8 \pi^3 f^2 N_A} \right) \frac{\epsilon_2(v)}{v}$

$$k_{rt} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |H_{21}|^2 \rho_{s2}(E_1) \rho_{s1}(E_1) dE_1$$

$$\begin{aligned}
 k_{rt} &= \frac{2\pi}{\hbar} \left\{ \frac{f^4}{n^4} D_{ba(1)} D_{ba(2)} \kappa^2 |R_{21}|^{-6} \right\} \int X_1(E) \rho_1(v) X_2(E) \rho_2(v) dE \\
 &= \frac{2\pi f^4}{\hbar n^4} \kappa^2 |R_{21}|^{-6} \left(\frac{3000 \ln 10 n h c}{8 \pi^3 f^2 N_A h} \right) \left(\frac{3 \hbar c^3 \Phi}{32 \pi^3 n f^2 \tau} \right) \frac{\int F_1(v) \epsilon_2(v) v^{-4} dv}{\int F_1(v) dv}
 \end{aligned}$$

Point-dipole approximation

12

FRET parameters from experiment

Finding R_0 : from experimental data

$$k_{rt} = \frac{2\pi}{\hbar} \frac{f^4}{n^4} \kappa^2 |R_{21}|^{-6} \left(\frac{3000 \ln 10 n \hbar c}{8\pi^3 f^2 N_A h} \right) \left(\frac{3\hbar c^3 \Phi}{32\pi^3 n f^2 \tau} \right) \frac{\int F_1(\nu) \epsilon_2(\nu) \nu^{-4} d\nu}{\int F_1(\nu) d\nu}$$

Combine numerical constants:

$$k_{rt} = \left(\frac{9000 \ln 10 \kappa^2 c^4 \Phi}{128 \pi^5 n^4 N_A \tau} \right) |R_{21}|^{-6} J$$

Units:

R_{21} – cm

ν – s^{-1} (Hz)

ϵ_2 – $M^{-1}cm^{-1}$

k_{tr} – 1/s

Overlap integral:

$$J = \frac{\int F_1(\nu) \epsilon_2(\nu) \nu^{-4} d\nu}{\int F_1(\nu) d\nu}$$

13

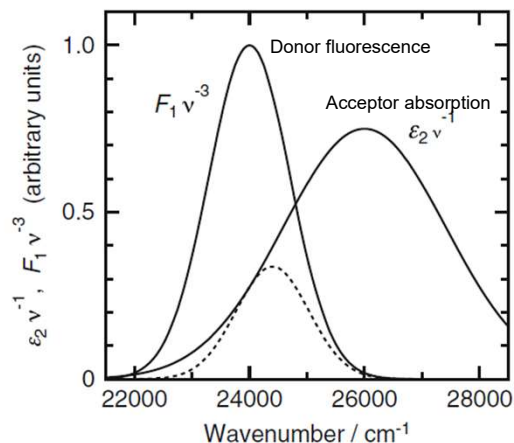
FRET parameters from experiment

Finding R_0 : from experimental data

$$k_{rt} = \left(\frac{9000 \ln 10 \kappa^2 c^4 \Phi}{128 \pi^5 n^4 N_A \tau} \right) |R_{21}|^{-6} J$$

$$J = \frac{\int F_1(\nu) \epsilon_2(\nu) \nu^{-4} d\nu}{\int F_1(\nu) d\nu}$$

Example:



14

FRET parameters from experiment

Finding R_0 : from experimental data

Can rewrite for different units:

$$\bar{J} = \frac{\int_{\text{Wavenumber scale (cm}^{-1}\text{)}} F_1(\bar{\nu}) \epsilon_2(\bar{\nu}) \bar{\nu}^{-4} d\bar{\nu}}{\int F_1(\bar{\nu}) d\bar{\nu}} = \frac{\int_{\text{Wavelength scale (cm)}} F_1(\lambda) \epsilon_2(\lambda) \lambda^2 d\lambda}{\int F_1(\lambda) \lambda^{-2} d\lambda}$$

$$k_{rt} = \left(\frac{8.78 \times 10^{23} \kappa^2 \Phi}{n^4 \tau} \right) |R_{21}|^{-6} \bar{J} s^{-1}$$

FRET efficiency

$$E = \frac{1}{(R_{21}/R_0)^6 + 1}$$

Where R_{21} is in Å

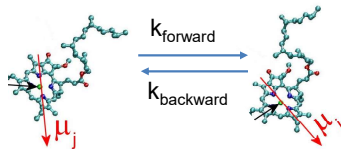
Compare with: $k_{rt} = \tau^{-1} (|R_{21}|/R_0)^{-6}$

$$R_0 = 9.80 \times 10^3 (\kappa^2 \Phi n^{-4} \bar{J})^{1/6} \text{Å} \quad \leftarrow \text{Förster radius:}$$

15

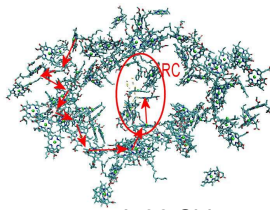
FRET: directionality

For identical chromophores energy transfer between molecules are the same.



If rates are equal $\rightarrow$ no preferable energy transfer direction

More chromophores: energy transfer is essentially a random walk for equivalent chromophores.



Getting to the destination (RC) is accidental and slow.

Speed up: tune transition energies for directional energy transfer

Photosystem I: 96 Chl a molecules

16

FRET: directionality

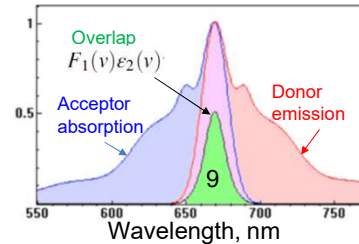
Tuning transition energies:
Asymmetry in overlap integrals

$$J = \frac{\int F_1(\nu)\epsilon_2(\nu)\nu^{-4}d\nu}{\int F_1(\nu)d\nu}$$

$$k_{rt} = \left(\frac{9000 \ln 10 \kappa^2 c^4 \Phi}{128 \pi^5 n^4 N_A \tau} \right) |R_{21}|^{-6} J$$

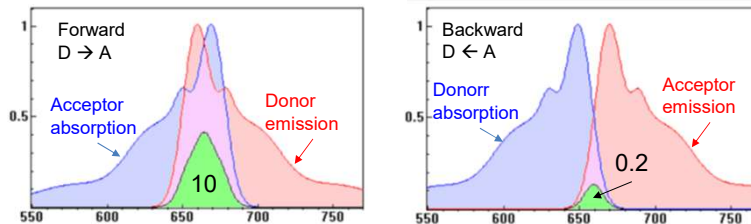
Identical pigments:

$$k_{forward} = k_{backward}$$



Acceptor is shifted to red (lower energy) by 10 nm in respect to donor:

$$k_{forward} \gg k_{backward}$$



17

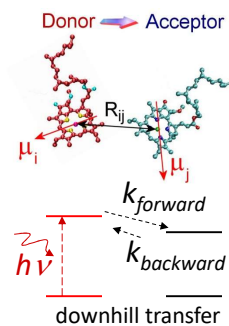
Thermodynamics of FRET

FRET obeys the laws of thermodynamics
(of course!)

Energy transfer may occur in both directions; the equilibrium constant is given by the Boltzmann distribution:

$$\frac{N_{donor}}{N_{acceptor}} = \frac{k_{backward}}{k_{forward}} = e^{\frac{\Delta E}{kT}}$$

A biologist usually writes $K_{eq} = 10^{\frac{\Delta G^\circ}{\ln 2 RT}}$

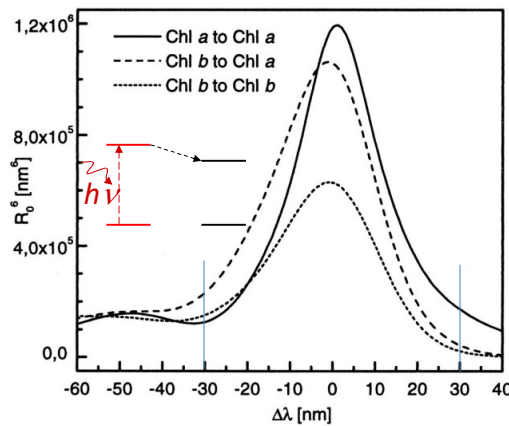


To ensure the directionality of energy flow the energy states of the acceptor state must be below that of the donor.

Downhill energy transfer - extra energy will be converted to vibrational energy of the donor - heat

18

Förster radius: Chl molecule



<http://www.biophysj.org/cgi/content/full/79/4/1706>
G. Cinque et al., Biophys J. 79 (2000) 1706

Note: spectra of Chl depend somewhat on their environment (nearby protein residues or other molecules)

$$k_{ij} = \frac{\kappa^2}{\tau_D} \left(\frac{R_0}{R_{ij}} \right)^6$$

$$\tau_{D, \text{Chl}} \approx 5 \text{ ns}$$

R_0 as a function of spectral shift between donor fluorescence and acceptor absorption

Förster radius for optimal spectral overlap for Chl a molecules is

$$(1.2 \times 10^6)^{1/6} \text{ nm} = 10 \text{ nm}$$

At this distance, the efficiency of excitation transfer is 50% (assuming $\kappa^2=1$)

19

Modeling energy transfer kinetics

Example: a system of 4 interacting pigments (1,2,3,4) where one of the pigments (4) serves as both the reaction center and antenna pigment.

Denote k_{ij} – energy transfer rates between the pigments $j \rightarrow i$
 τ_i – excited state lifetime of pigments $i=1,2,3,4$
 $n_i(t)$ – time-dependent population of excited pigments
 k_{54} – charge separation time (assume irreversible)

Can write down a system of kinetic equations:

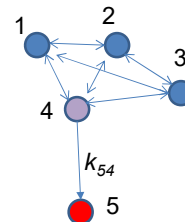
$$\frac{dn_1}{dt} = - \left(\frac{1}{\tau_1} + k_{21} + k_{31} + k_{41} \right) n_1 + k_{12}n_2 + k_{13}n_3 + k_{14}n_4$$

$$\frac{dn_2}{dt} = - \left(\frac{1}{\tau_2} + k_{12} + k_{32} + k_{42} \right) n_2 + k_{21}n_1 + k_{23}n_3 + k_{24}n_4$$

$$\frac{dn_3}{dt} = \dots$$

$$\frac{dn_4}{dt} = - \left(\frac{1}{\tau_4} + k_{14} + k_{24} + k_{34} + k_{54} \right) n_4 + k_{41}n_1 + k_{42}n_2 + k_{43}n_3$$

$$\frac{dn_5}{dt} = k_{54}n_4$$

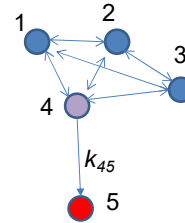


Note: assumed fast transfer, ignored radiative and nonradiative decay channels
 (can be included as, for example, for the first equation, adding term $-k_{11}n_1$, with k_{11} rate of decay)

20

Modeling energy transfer kinetics

$$\begin{aligned}\frac{dn_1}{dt} &= -\left(\frac{1}{\tau_1} + k_{21} + k_{31} + k_{41}\right)n_1 + k_{12}n_2 + k_{13}n_3 + k_{14}n_4 \\ \frac{dn_2}{dt} &= -\left(\frac{1}{\tau_2} + k_{12} + k_{32} + k_{42}\right)n_2 + k_{21}n_1 + k_{23}n_3 + k_{24}n_4 \\ \frac{dn_3}{dt} &= \dots \\ \frac{dn_4}{dt} &= -\left(\frac{1}{\tau_4} + k_{14} + k_{24} + k_{34} + k_{54}\right)n_4 + k_{41}n_1 + k_{42}n_2 + k_{43}n_3 \\ \frac{dn_5}{dt} &= k_{54}n_4\end{aligned}$$



Need: initial populations of all states: $n_i(0)$

Can solve for population dynamics

? How can we find the energy transfer efficiency?

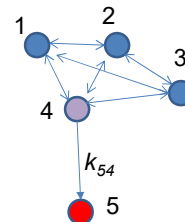
→ can investigate $n_5(t)$ at $t \rightarrow \infty$ and compare with total initial population of excited states

21

Modeling energy: computational

$$\begin{aligned}\frac{dn_1}{dt} &= -(k_{11} + k_{21} + k_{31} + k_{41})n_1 + k_{12}n_2 + k_{13}n_3 + k_{14}n_4 \\ \frac{dn_2}{dt} &= -(k_{12} + k_{22} + k_{32} + k_{42})n_2 + k_{21}n_1 + k_{23}n_3 + k_{24}n_4 \\ \frac{dn_3}{dt} &= \dots \\ \frac{dn_4}{dt} &= -(k_{14} + k_{24} + k_{34} + k_{44} + k_{54})n_4 + k_{41}n_1 + k_{42}n_2 + k_{43}n_3 \\ \frac{dn_5}{dt} &= n_4 k_{54}\end{aligned}$$

Where $k_{ii} = \frac{1}{\tau_i}$



Notice: can use matrices: $\frac{d}{dt} \mathbf{n} = \mathbf{K} \mathbf{n}$

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ \dots \\ n_n \end{pmatrix} \quad \mathbf{K} = \begin{pmatrix} -(k_{11} + k_{21} + \dots) & k_{12} & \dots & k_{1n} \\ k_{21} & -(k_{12} + k_{22} + \dots) & \dots & k_{2n} \\ \dots & \dots & \dots & \dots \\ k_{n1} & k_{n2} & \dots & -(k_{1n} + k_{2n} + \dots) \end{pmatrix}$$

22

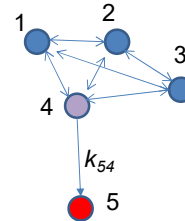
Modeling energy: computational

Notice: can use matrices: $\frac{d}{dt}\mathbf{n} = \mathbf{R}\mathbf{n}$

Solving using a computer: recurrently, using $d\mathbf{n} = \mathbf{R}\mathbf{n}dt$

1. Pick a short time interval Δt
2. Define initial conditions, vector $\mathbf{n}(t=0)$
3. Recurrently calculate $\mathbf{n}$ at the next time

$$\mathbf{n}(t + \Delta t) = \mathbf{n}(t) + \Delta \mathbf{n}(t) = \mathbf{n}(t) + \mathbf{R}\mathbf{n}(t)\Delta t$$



23

Modeling energy: computational

Python code: assume N pigments + one trap, i.e. N+1 rate equations

```
import numpy as np # imports numerical python library
```

Building rate matrix:

$$\mathbf{R} = \begin{pmatrix} -\sum_j k_{j1} & k_{12} & \cdots & k_{1n} \\ k_{21} & -\sum_j k_{j2} & \cdots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \cdots & -\sum_j k_{jn} \end{pmatrix}$$

1. Define K matrix:

$$\mathbf{K} = \begin{pmatrix} k_{11} & k_{12} & \cdots & k_{1n} \\ k_{21} & k_{22} & \cdots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \cdots & k_{nn} \end{pmatrix}$$

forward

backward

$$\text{Diagonal: } k_{ii} = \frac{1}{\tau_i} \text{ (lifetime)}$$

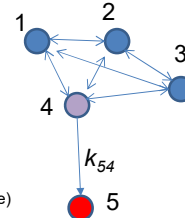
```
N = 5; N1=N+1 # define the size of the system
K = np.zeros((N1,N1),dtype=float) # create N1xN1 matrix of type float with K13=0
# define all non-zero elements, like K[i][j] = whatever_the_value
```

2. Find diagonal elements of $\mathbf{R}$: $\mathbf{D} = -\mathbf{I}_1\mathbf{K}$, where $\mathbf{I}_1 = \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}$

```
D = -np.ones((N1)) @ K # result is column matrix, @ means multiply by matrix rules
```

3. Replace the diagonal in the $\mathbf{K}$ with $\mathbf{D}$ to obtain $\mathbf{R}$

```
R = K.copy() # make a copy of K
np.fill_diagonal(R,D.T) # transpose D to make it a row and set diagonal in R to it
```



24

Modeling energy: computational

Python code: assume N pigments + one trap trap, i.e., $N_1=N+1$ rate equations

```
import numpy as np # imports numerical python library
N = 5; N1=N+1 # define the size of the system
K = np.zeros((N1,N1),dtype=float) # create N1xN1 matrix of type float with  $K_{ij}=0$ 
# define all non-zero elements, like  $K[i][j]$  = whatever_the_value, depends on system
D = -np.ones((N1)) @ K # result is column matrix, @ means multiply by matrix rules
R = K.copy() # make a copy of K
np.fill_diagonal(R,D.T) # transpose D to make it a row and set diagonal in R to it
```

4. Define $n(t=0)$

```
n = np.ones((N))/(N-1) # make array of state populations, equal populations, nnormalized
n[N] = 0 # no initial population on trap
N = n.T # transpose to make a row vector into a column vector
```

5. Calculate population dynamics

```
dynam = [] # reserve list to store n at different times
tt = [] # list of respective times
tmax, dt = 2000, 0.01 # calculate all the way to tmax with step dt
t_record = 100 # store only every t_record point
t_point = 0
for t in np.arange(0.,tmax,dt):
    if t_point == 0: dynam.append(n); tt.append(t); t_point = t_record
    t_point -= 1
    n = n + R*n*dt

tt = np.array(tt) # make list into array
dynam = np.array(dynam).T # transpose,
# dynam[N][-1] is population of trap at the end of simulation
```

25

Modeling energy: computational

Python code: assume N pigments + one trap trap, i.e., $N_1=N+1$ rate equations

6. Plot population dynamics

```
import matplotlib.pyplot as plt # import plotting library

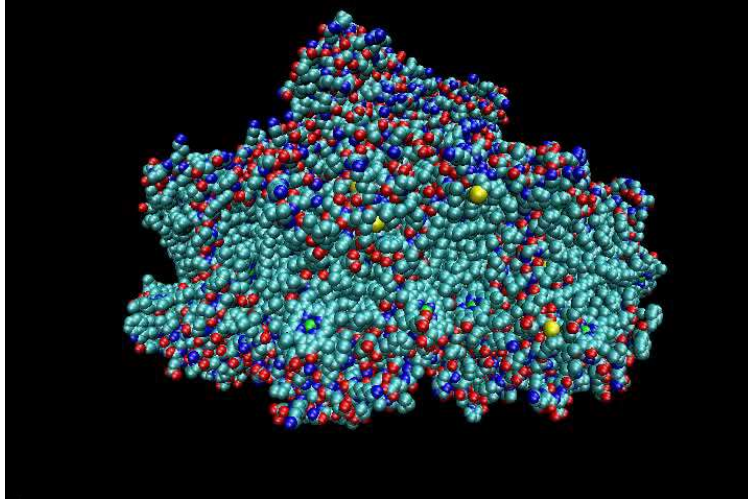
fig, (ax1,ax2) = plt.subplots(2,1, figsize = (5,5), sharex=True) # make 2 graphs

ax1.plot(tt,dynam[0]) # plot population of pigment 0 and 1 in graph 1
ax1.plot(tt,dynam[1])
ax2.plot(tt,dynam[N]) # plot population of the trap in second graph
ax2.plot(tt,np.sum(dynam[:-1],axis=0)) # add populations of antenna pigments and plot
print(f'Quantum yield: {dynam[N][-1]}') # last element - quantum efficiency
```

26

FRET in complex systems: PS I

Example: photosystem I



27

Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

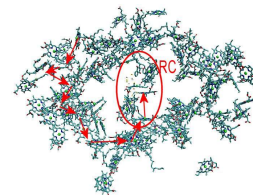
Solve System of 96 rate equations:

$$\begin{cases} \frac{dn_1}{dt} = -\left(\frac{1}{\tau_{1f}} + k_{21} + k_{31} + \dots + k_{n1}\right)n_1 + \overset{\substack{\text{Transfer rate from} \\ \text{molecule 3 to 1}}}{k_{13}n_3} + \dots + k_{1n}n_n \\ \frac{dn_2}{dt} = -\left(\frac{1}{\tau_{2f}} + k_{12} + k_{32} + \dots + k_{n2}\right)n_2 + k_{21}n_1 + k_{23}n_3 + \dots + k_{2n}n_n \\ \dots \end{cases}$$

In principle can be solved analytically, but the resulting equation will be much too long...

Solve on computer:

1. Set initially excited molecule (k) population to N_k to 1
Leave all others $N_{i \neq k} = 0$. Time $t=0$
2. Increase time by a small amount dt . Use the equations above to find changes in population dN_i after time dt
3. Repeat step 2 until needed to generate $N_i(t)$ profiles
4. Assuming spectra for each excited state are known – generate $S(\lambda, t)$ and compare with the experiment



28

Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

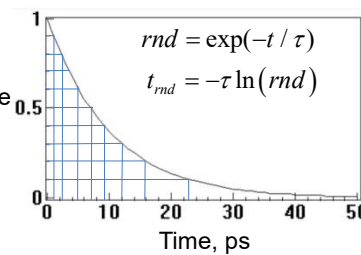
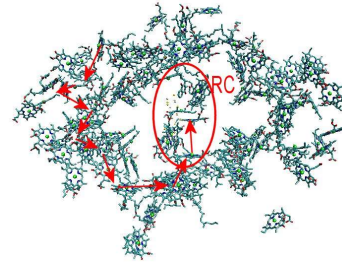
Alternative: **Monte Carlo**

1. Define which molecule gets excited.
Can be based on the absorption spectra of each pigment and defined photon wavelength
2. Weighted on k_{ij} generate "random" transfer time τ_{ij} to every other molecule in the system.
3. Pick the shortest τ_{ij} , record that the excitation "hopped" to that pigment j at time $t + \tau_{ij}$
4. Go to step 2 to see where and when the excitation "hops" next

What you need: a random number generator that produces an exponential distribution of events in time

How can one do that?

Generate random number rnd between (0,1)
Project to exponent, find respective time t_{rnd}



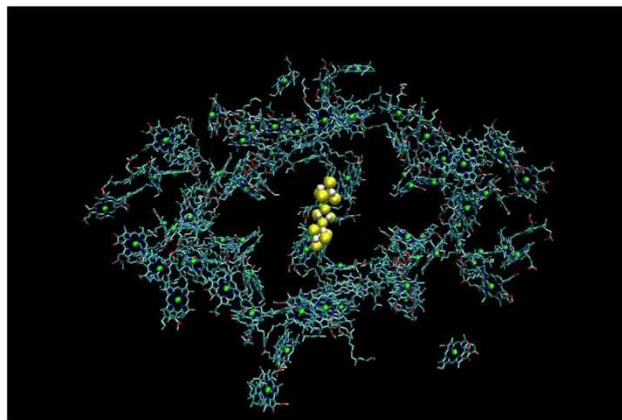
29

Analyzing FRET kinetics data

Example: photosystem I: 96 pigments, 96x96 energy transfer rates

Alternative: **Monte Carlo**

Time slowed down a million million times! $1\text{ps} = 10^{-12}\text{s} \rightarrow 1\text{s}$



30